

1. In an experiment with vernier callipers of least count 0.1 mm, when two jaws are joined together the zero of vernier scale lies right to the zero of the main scale and 6th division of vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 cm and 3.3 cm marks and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as [2023 (10 Apr Shift 2)]
- (1) 3.22 cm
(2) 3.18 cm
(3) 3.26 cm
(4) 3.25 cm

ANSWER KEYS

1. (2)

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The least count is $LC = 0.1 \text{ mm}$

Zero error (ZE) = $6 \times LC = 6 \times 0.1 \text{ mm} = 0.6 \text{ mm}$

The total reading is

$$TR = MSR + (VSR \times LC) - ZE$$

$$\Rightarrow TR = 32 \text{ mm} + (0.1 \times 4) \text{ mm} - 0.6 \text{ mm}$$

$$\Rightarrow TR = 31.8 \text{ mm} = 3.18 \text{ cm}$$